第4章 不定积分

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§2 换元积分法

2.1 第一类换元法(凑微分法)

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2. 1 第一换元法(凑微分法)

问题 $\int \cos 2x dx \overset{?}{=} \sin 2x + C,$

解决方法 利用一阶微分形式不变性

$$\frac{1}{2} \cdot d(\sin 2x) = \frac{1}{2} \cdot \cos 2x d(2x)$$

$$\Downarrow$$

$$d\left(\frac{1}{2} \cdot \sin 2x\right) = \cos 2x dx$$

$$\int \cos 2x dx = \frac{1}{2} \sin 2x + C$$

$\because dF(x) = f(x)dx; \quad dF(u) = f(u)du$

自变量　　　　中间变量

$\therefore \int f(x)dx = F(x) + C; \quad \int f(u)du = F(u) + C.$

$u = \varphi(x)$

定理1（第一换元法）设 $\int f(u)du = F(u) + C,$

且 $u = \varphi(x)$ 可微, 则

$$\int f[\varphi(x)]\varphi'(x)dx = F[\varphi(x)] + C$$

证

定理1（第一换元法）设 $\int f(u)du = F(u) + C$,

且 $u = \varphi(x)$ 可微, 则

$$\int f[\varphi(x)]\varphi'(x)dx = F[\varphi(x)] + C$$

可形象地表述为

$$\int f[\varphi(x)]\varphi'(x)dx \overset{\text{凑微分}}{=} \int f[\varphi(x)]d\varphi(x) \overset{\varphi(x)=u}{=} \int f(u)du$$

$$= F(u) + C \overset{u=\varphi(x)}{=} F[\varphi(x)] + C$$

这种积分方法称为第一换元公式（**凑微分法**）

说明 使用此公式的关键在于将 $\int g(x)dx$ 化为 $\int f[\varphi(x)]\varphi'(x)dx.$

观察重点不同，所得结论不同.

第一类换元法解决的问题

$$\int f[\varphi(x)]\varphi'(x)\mathrm{d}x = \left.\int f(u)\mathrm{d}u\right|_{u=\varphi(x)}$$

难求　　　　　　易求

以下是最基本且经常会遇到的结果:

$$\int e^{\varphi(x)}\varphi'(x)\mathrm{d}x=\int e^{\varphi(x)}\mathrm{d}\varphi(x)=e^{\varphi(x)}+C$$

$$\int \varphi'(x)\cos\varphi(x)\mathrm{d}x=\int \cos\varphi(x)\mathrm{d}\varphi(x)=\sin\varphi(x)+C$$

$$\int \varphi'(x)\sin\varphi(x)\mathrm{d}x=\int \sin\varphi(x)\mathrm{d}\varphi(x)=-\cos\varphi(x)+C$$

$$\int \varphi(x)\varphi'(x)\mathrm{d}x=\int \varphi(x)\mathrm{d}\varphi(x)=\frac{1}{2}\varphi^2(x)+C$$

$$\int \frac{\varphi'(x)}{\varphi^2(x)}\mathrm{d}x=\int \frac{1}{\varphi^2(x)}\mathrm{d}\varphi(x)=-\frac{1}{\varphi(x)}+C$$

$$\int \frac{\varphi'(x)}{\varphi(x)}\mathrm{d}x=\int \frac{1}{\varphi(x)}\mathrm{d}\varphi(x)=\ln|\varphi(x)|+C$$

例1　求 $\int \sin 2x dx.$

解（一）$\int \sin 2x dx = \frac{1}{2}\int \sin 2x d(2x)$

$= -\frac{1}{2}\cos 2x + C;$

解（二）$\int \sin 2x dx = 2\int \sin x \cos x dx$

$= 2\int \sin x d(\sin x) = (\sin x)^2 + C;$

解（三）$\int \sin 2x dx = 2\int \sin x \cos x dx$

$= -2\int \cos x d(\cos x) = -(\cos x)^2 + C.$

例2 求$\int\frac{1}{3+2x}dx.$

解 $$\frac{1}{3+2x}=\frac{1}{2}\cdot\frac{1}{3+2x}\cdot(3+2x)',$$

$$\int\frac{1}{3+2x}dx=\frac{1}{2}\int\frac{1}{3+2x}\cdot(3+2x)'dx$$

$$=\frac{1}{2}\int\frac{1}{u}du=\frac{1}{2}\ln|u|+C=\frac{1}{2}\ln|3+2x|+C.$$

一般地 $$\int f(ax+b)dx=\frac{1}{a}\left[\int f(u)du\right]_{u=ax+b}$$

例 求 $\int (ax+b)^m \,\mathrm{d}\, x \ \ (a \neq 0, m \neq -1).$

解: 令 $u = ax + b$,则 $\mathrm{d}\,u = a\mathrm{d}x$, 故

$$\text{原式} = \int u^m \ \frac{1}{a} \mathrm{d}\,u = \frac{1}{a} \cdot \frac{1}{m+1} u^{m+1} + C$$

$$= \frac{1}{a(m+1)} (ax+b)^{m+1} + C$$

注 当 $m = -1$ 时

$$\int \frac{\mathrm{d}\,x}{ax+b} = \frac{1}{a} \ln|ax+b| + C$$

例3 求 $\int \frac{dx}{x^2-a^2}$

解：原式 $=\frac{1}{2a}\int(\frac{1}{x-a}-\frac{1}{x+a})dx$

$$=\frac{1}{2a}[\int\frac{d(x-a)}{x-a}-\int\frac{d(x+a)}{x+a}]$$

$$=\frac{1}{2a}\left[\ln|x-a|-\ln|x+a|\right]+c$$

$$=\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|+c$$

例4 求 $\int \cos 3x \cos 2x dx$.

解 $\cos A \cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$,

$$\cos 3x \cos 2x = \frac{1}{2}(\cos x + \cos 5x),$$

$$\int \cos 3x \cos 2x dx = \frac{1}{2}\int (\cos x + \cos 5x) dx$$

$$= \frac{1}{2}\sin x + \frac{1}{10}\sin 5x + C.$$

例5 求$\int \tan x \mathrm{d}x$.

解 $\int \tan x \mathrm{d}x = \int \frac{\sin x}{\cos x} \mathrm{d}x = -\int \frac{\mathrm{d}\cos x}{\cos x}$

$= -\ln|\cos x| + C$

类似

$\int \cot x \mathrm{d}x = \int \frac{\cos x \mathrm{d}x}{\sin x} = \int \frac{\mathrm{d}\sin x}{\sin x}$

$= \ln|\sin x| + C$

例6 求 $\int \frac{1}{x(1+2\ln x)}dx.$

解 $\int \frac{1}{x(1+2\ln x)}dx = \int \frac{1}{1+2\ln x}d(\ln x)$

$= \frac{1}{2}\int \frac{1}{1+2\ln x}d(1+2\ln x)$

$u = 1+2\ln x$

$= \frac{1}{2}\int \frac{1}{u}du = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln|1+2\ln x| + C.$

例7 求 $\int \frac{\mathrm{d}x}{a^2+x^2}$.

解: $\int \frac{\mathrm{d}x}{a^2+x^2} = \frac{1}{a^2}\int \frac{\mathrm{d}x}{1+(\frac{x}{a})^2}$

令 $u=\frac{x}{a}$, 则 $\mathrm{d}u=\frac{1}{a}\mathrm{d}x$

$= \frac{1}{a}\int \frac{\mathrm{d}u}{1+u^2} = \frac{1}{a}\arctan u + C$

$= \frac{1}{a}\arctan(\frac{x}{a}) + C$

想到公式

$$\int \frac{\mathrm{d}u}{1+u^2} = \arctan u + C$$

例8 求 $\int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}}\ (a>0).$

解: $\int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}} = \int \frac{\mathrm{d}x}{a\sqrt{1-(\frac{x}{a})^2}} = \int \frac{\mathrm{d}(\frac{x}{a})}{\sqrt{1-(\frac{x}{a})^2}}$

$$= \arcsin\frac{x}{a} + C$$

想到 $\int \frac{\mathrm{d}u}{\sqrt{1-u^2}} = \arcsin u + C$

$$\int f[\varphi(x)]\varphi'(x)\mathrm{d}x = \int f(\varphi(x))\mathrm{d}\varphi(x)$$ **(直接配元)**

例9　求$\int \frac{1}{x^2-8x+25}dx.$

解　$\int \frac{1}{x^2-8x+25}dx = \int \frac{1}{(x-4)^2+9}dx$

$$=\frac{1}{3^2}\int \frac{1}{\left(\frac{x-4}{3}\right)^2+1}dx = \frac{1}{3}\int \frac{1}{\left(\frac{x-4}{3}\right)^2+1}d\left(\frac{x-4}{3}\right)$$

$$=\frac{1}{3}\arctan\frac{x-4}{3}+C.$$

例10 求 $\int \frac{1}{1+e^x}dx.$

解 $\int \frac{1}{1+e^x}dx = \int \frac{1+e^x-e^x}{1+e^x}dx$

$= \int \left(1-\frac{e^x}{1+e^x}\right)dx = \int dx - \int \frac{e^x}{1+e^x}dx$

$= \int dx - \int \frac{1}{1+e^x}d(1+e^x)$

$= x - \ln(1+e^x) + C.$

例11 求$\int (1-\frac{1}{x^2})e^{x+\frac{1}{x}}dx.$

解 $\because \left(x+\frac{1}{x}\right)' = 1-\frac{1}{x^2},$

$$\therefore \int (1-\frac{1}{x^2})e^{x+\frac{1}{x}}dx$$

$$=\int e^{x+\frac{1}{x}}d(x+\frac{1}{x}) = e^{x+\frac{1}{x}}+C.$$

例12 求 $\int \frac{1}{1+\cos x} dx.$

解 $$\int \frac{1}{1+\cos x} dx = \int \frac{1-\cos x}{(1+\cos x)(1-\cos x)} dx$$

$$= \int \frac{1-\cos x}{1-\cos^2 x} dx = \int \frac{1-\cos x}{\sin^2 x} dx$$

$$= \int \frac{1}{\sin^2 x} dx - \int \frac{1}{\sin^2 x} d(\sin x)$$

$$= -\cot x + \frac{1}{\sin x} + C.$$

例13 求$\int \csc x dx$.

解（一）$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{1}{2\sin\frac{x}{2}\cos\frac{x}{2}} dx$

$$= \int \frac{1}{\tan\frac{x}{2}\left(\cos\frac{x}{2}\right)^2} d\left(\frac{x}{2}\right) = \int \frac{1}{\tan\frac{x}{2}} d\left(\tan\frac{x}{2}\right)$$

$$= \ln\left|\tan\frac{x}{2}\right| + C = \ln\left|\csc x - \cot x\right| + C.$$

$$注:\tan\frac{x}{2}=\frac{\sin\frac{x}{2}}{\cos\frac{x}{2}}=\frac{2\sin\frac{x}{2}\sin\frac{x}{2}}{2\cos\frac{x}{2}\sin\frac{x}{2}}$$

$$=\frac{1-\cos x}{\sin x}=\csc x-\cot x$$

$$\int\csc x dx=\ln\left|\csc x-\cot x\right|+C.$$

$$=-\ln\left|\csc x+\cot x\right|+C.$$

解（二）$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{\sin x}{\sin^2 x} dx$

$$= -\int \frac{1}{1-\cos^2 x} d(\cos x) \qquad u = \cos x$$

$$= -\int \frac{1}{1-u^2} du = -\frac{1}{2} \int \left(\frac{1}{1-u} + \frac{1}{1+u} \right) du$$

$$= \frac{1}{2} \ln \left| \frac{1-u}{1+u} \right| + C = \frac{1}{2} \ln \left| \frac{1-\cos x}{1+\cos x} \right| + C.$$

类似地可推出 $\int \sec x dx = \ln \left| \sec x + \tan x \right| + C.$

常用的几种凑微分形式:

(1) $\int f(ax+b)\mathrm{d}x = \frac{1}{a}\int f(ax+b)\ \mathrm{d}(ax+b)$

(2) $\int f(x^n)x^{n-1}\,\mathrm{d}x = \frac{1}{n}\int f(x^n)\ \mathrm{d}x^n$

(3) $\int f(x^n)\frac{1}{x}\mathrm{d}x = \frac{1}{n}\int f(x^n)\ \frac{1}{x^n}\ \mathrm{d}x^n$

万能凑幂法

(4) $\int f(\sin x)\cos x\mathrm{d}x = \int f(\sin x)\,\mathrm{d}\sin x$

(5) $\int f(\cos x)\sin x\mathrm{d}x = -\int f(\cos x)\ \mathrm{d}\cos x$

$$(6)\ \int f(\tan x)\sec^2 x\mathrm{d}x = \int f(\tan x)\ \mathrm{d}\tan x$$

$$(7)\ \int f(e^x)e^x\mathrm{d}x = \int f(e^x)\ \mathrm{d}e^x$$

$$(8)\ \int f(\ln x)\frac{1}{x}\mathrm{d}x = \int f(\ln x)\ \mathrm{d}\ln x$$

问题 $\int x^5\sqrt{1-x^2}\,dx = ?$

解决方法 改变中间变量的设置方法.

过程 令 $x=\sin t \Rightarrow dx=\cos t\,dt$,

$$\int x^5\sqrt{1-x^2}\,dx=\int(\sin t)^5\sqrt{1-\sin^2 t}\cos t\,dt$$

$$=\int\sin^5 t\cos^2 t\,dt \quad = \cdots\cdots$$

（应用“凑微分”即可求出结果）

2.2 换元法则(II)----第二类换元法

第一类换元法解决的问题

$$\underset{\text{难求}}{\int f[\varphi(x)]\varphi'(x)\mathrm{d}x} = \underset{\text{易求}}{\int f(u)\mathrm{d}u}\Big|_{u=\varphi(x)}$$

若所求积分 $\int f(u)\mathrm{d}u$ 难求，

$\int f[\varphi(x)]\varphi'(x)\mathrm{d}x$ 易求，

则得第二类换元积分法 .

定理2 设 $x=\psi(t)$ 是单调可导函数，且 $\psi'(t)\neq 0$，$f[\psi(t)]\psi'(t)$ 具有原函数，则有换元公式

$$\int f(x)dx = \left[\int f[\psi(t)]\psi'(t)dt\right]_{t=\psi^{-1}(x)}$$

其中 $t=\psi^{-1}(x)$ 是 $x=\psi(t)$ 的反函数 .

证: 设 $f[\psi(t)]\psi'(t)$ 的原函数为 $\Phi(t)$，令

$$F(x)=\Phi[\psi^{-1}(x)]$$

则

$$F'(x)=\frac{\mathrm{d}\Phi}{\mathrm{d}t}\cdot\frac{\mathrm{d}t}{\mathrm{d}x}=f[\psi(t)]\cancel{\psi'(t)}\cdot\frac{1}{\cancel{\psi'(t)}}=f(x)$$

$$\therefore \int f(x)\mathrm{d}\,x=F(x)+C=\Phi[\psi^{-1}(x)]+C$$

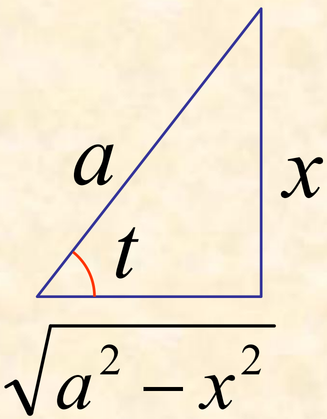
$$=\int f[\psi(t)]\psi'(t)\,\mathrm{d}\,t\Big|_{t=\psi^{-1}(x)}$$

例1 求 $\int \sqrt{a^2 - x^2}\,\mathrm{d}x \quad (a > 0)$.

解: 令 $x = a\sin t,\ t \in (-\frac{\pi}{2}, \frac{\pi}{2})$, 则

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2 t} = a\cos t$$

$$\mathrm{d}x = a\cos t\,\mathrm{d}t$$



$\therefore$ 原式 $= \int a\cos t \cdot a\cos t\,\mathrm{d}t = a^2 \int \cos^2 t\,\mathrm{d}t$

$$= a^2\left(\frac{t}{2} + \frac{\sin 2t}{4}\right) + C$$

$$\sin 2t = 2\sin t\cos t = 2 \cdot \frac{x}{a} \cdot \frac{\sqrt{a^2 - x^2}}{a}$$

$$= \frac{a^2}{2}\arcsin\frac{x}{a} + \frac{1}{2}x\sqrt{a^2 - x^2} + C$$

例2 求 $\int x^3\sqrt{4-x^2}dx.$

解 令 $x=2\sin t \quad dx=2\cos t dt \quad t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

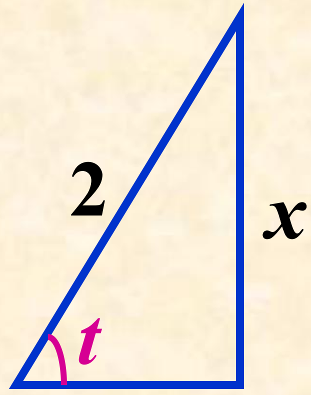
$$\int x^3\sqrt{4-x^2}dx=\int(2\sin t)^3\sqrt{4-4\sin^2 t}\cdot 2\cos t dt$$

$$=32\int\sin^3 t\cos^2 t dt=32\int\sin t(1-\cos^2 t)\cos^2 t dt$$

$$=-32\int(\cos^2 t-\cos^4 t)d\cos t$$

$$=-32\left(\frac{1}{3}\cos^3 t-\frac{1}{5}\cos^5 t\right)+C$$

$$=-\frac{4}{3}\left(\sqrt{4-x^2}\right)^3+\frac{1}{5}\left(\sqrt{4-x^2}\right)^5+C.$$



$\sqrt{4-x^2}$

例3 求 $\int \frac{dx}{\sqrt{x^2+a^2}} \quad (a>0)$.

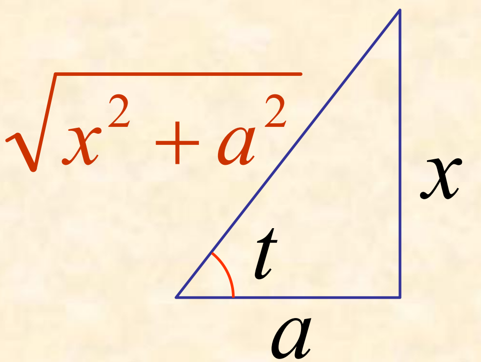
解: 令 $x = a\tan t\,,\ t \in (-\frac{\pi}{2},\frac{\pi}{2})$, 则

$$\sqrt{x^2+a^2} = \sqrt{a^2\tan^2 t + a^2} = a\sec t$$

$$dx = a\sec^2 t\,dt$$

$\therefore$ 原式 $= \int \frac{a\sec^2 t}{a\sec t}\,dt = \int \sec t\,dt$

$$= \ln\left|\sec t + \tan t\right| + C_1$$



$$= \ln\left[\frac{\sqrt{x^2+a^2}}{a} + \frac{x}{a}\right] + C_1$$

$$= \ln\left[x + \sqrt{x^2+a^2}\right] + C \qquad (C = C_1 - \ln a)$$

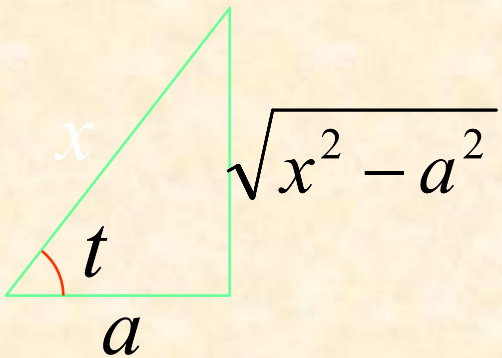
例4 求 $\int \frac{\mathrm{d}x}{\sqrt{x^2 - a^2}} \quad (a > 0).$

解: 当$x > a$时,令 $x = a\sec t,\ t \in (0, \frac{\pi}{2})$, 则

$$\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2 t - a^2} = a\tan t$$

$$\mathrm{d}x = a\sec t\tan t\,\mathrm{d}t$$

$$\therefore \text{原式} = \int \frac{a\sec t\tan t}{a\tan t}\mathrm{d}t = \int \sec t\,\mathrm{d}t$$



$$= \ln\left|\sec t + \tan t\right| + C_1$$

$$= \ln\left|\frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a}\right| + C_1$$

$$= \ln\left|x + \sqrt{x^2 - a^2}\right| + C \qquad (C = C_1 - \ln a)$$

当$x<-a$时，令 $x=-u$，则 $u>a$，于是

$$\int\frac{\mathrm{d}x}{\sqrt{x^2-a^2}}=-\int\frac{\mathrm{d}u}{\sqrt{u^2-a^2}}=-\ln\left|u+\sqrt{u^2-a^2}\right|+C_1$$

$$=-\ln\left|-x+\sqrt{x^2-a^2}\right|+C_1$$

$$=-\ln\left|\frac{a^2}{-x-\sqrt{x^2-a^2}}\right|+C_1$$

$$=\ln\left|x+\sqrt{x^2-a^2}\right|+C\quad(C=C_1-2\ln a)$$

$x>a$ 时，$\displaystyle\int\frac{\mathrm{d}x}{\sqrt{x^2-a^2}}=\ln\left|x+\sqrt{x^2-a^2}\right|+C$

说明(1) 以上几例所使用的均为三角代换.

三角代换的目的是化掉根式.

一般规律如下：当被积函数中含有

(1) $\sqrt{a^2-x^2}$ 可令 $x=a\sin t$;

(2) $\sqrt{a^2+x^2}$ 可令 $x=a\tan t$;

(3) $\sqrt{x^2-a^2}$ 可令 $x=a\sec t$.

说明(2)

被积函数含有 $\sqrt{x^2+a^2}$ 或 $\sqrt{x^2-a^2}$ 时, 除采用

三角代换外, 还可利用公式

$$\mathrm{ch}^2 t-\mathrm{sh}^2 t=1$$

采用双曲代换

$$x=a\,\mathrm{sh}\,t \quad \text{或} \quad x=a\,\mathrm{ch}\,t$$

消去根式, 所得结果一致.

说明(3) 当分母的阶较高时, 可采用倒代换 $x=\frac{1}{t}$.

例5 求 $\int \frac{1}{x(x^7+2)}dx$

解 令 $x=\frac{1}{t} \Rightarrow dx=-\frac{1}{t^2}dt$,

$$\int \frac{1}{x(x^7+2)}dx=\int \frac{t}{\left(\frac{1}{t}\right)^7+2}\cdot\left(-\frac{1}{t^2}\right)dt=-\int \frac{t^6}{1+2t^7}dt$$

$$=-\frac{1}{14}\ln|1+2t^7|+C=-\frac{1}{14}\ln|2+x^7|+\frac{1}{2}\ln|x|+C.$$

说明(4) 当被积函数含有两种或两种以上的根式$\sqrt[k]{x},\cdots,\sqrt[l]{x}$时，可采用令$x=t^n$（其中 n为各根指数的**最小公倍数**）

例6 求 $\displaystyle\int\frac{1}{\sqrt{x}(1+\sqrt[3]{x})}dx.$

解 令 $x=t^6\Rightarrow dx=6t^5dt,$

$$\int\frac{1}{\sqrt{x}(1+\sqrt[3]{x})}dx=\int\frac{6t^5}{t^3(1+t^2)}dt=\int\frac{6t^2}{1+t^2}dt$$

$$=6\int\left(1-\frac{1}{1+t^2}\right)dt=6[t-\arctan t]+C$$

$$=6[\sqrt[6]{x}-\arctan\sqrt[6]{x}]+C.$$

说明： 1. 第二类换元法常见类型：

(1) $\int f(x, \sqrt[n]{ax+b})\,\mathrm{d}x$, 令 $t = \sqrt[n]{ax+b}$

(2) $\int f\left(x, \sqrt[n]{\frac{ax+b}{cx+d}}\right)\mathrm{d}x$, 令 $t = \sqrt[n]{\frac{ax+b}{cx+d}}$

(3) $\int f(x, \sqrt{a^2 - x^2})\,\mathrm{d}x$, 令 $x = a\sin t$ 或 $x = a\cos t$

(4) $\int f(x, \sqrt{a^2 + x^2})\,\mathrm{d}x$, 令 $x = a\tan t$ 或 $x = a\,\mathrm{sh}\,t$

(5) $\int f(x, \sqrt{x^2 - a^2})\,\mathrm{d}x$, 令 $x = a\sec t$ 或 $x = a\,\mathrm{ch}\,t$

(6) $\int f(a^x)\,\mathrm{d}x$, $t = a^x$

(7) 分母中因子次数较高时, 可试用倒代换

说明(5) 积分中为了化掉根式是否一定采用三角代换（或双曲代换）并不是绝对的，需根据被积函数的情况来定.

例7 求 $\int \frac{x^5}{\sqrt{1+x^2}}dx$ （三角代换很繁琐）

解 令 $t=\sqrt{1+x^2} \Rightarrow x^2=t^2-1, \quad xdx=tdt,$

$$\int \frac{x^5}{\sqrt{1+x^2}}dx=\int \frac{\left(t^2-1\right)^2}{t}tdt=\int\left(t^4-2t^2+1\right)dt$$

$$=\frac{1}{5}t^5-\frac{2}{3}t^3+t+C=\frac{1}{15}(8-4x^2+3x^4)\sqrt{1+x^2}+C.$$

例8 求 $\int \frac{1}{\sqrt{1+e^x}}dx.$

解 令 $t=\sqrt{1+e^x} \Rightarrow e^x = t^2-1,$

$$x=\ln\left(t^2-1\right),\quad dx=\frac{2t}{t^2-1}dt,$$

$$\int \frac{1}{\sqrt{1+e^x}}dx = \int \frac{2}{t^2-1}dt = \int\left(\frac{1}{t-1}-\frac{1}{t+1}\right)dt$$

$$=\ln\left|\frac{t-1}{t+1}\right|+C = 2\ln\left(\sqrt{1+e^x}-1\right)-x+C.$$

（8）万能代换

令 $t=\tan\dfrac{x}{2}$　　$x=2\arctan t$　（万能代换公式）

$$\boxed{\sin x=\frac{2t}{1+t^2},\quad \cos x=\frac{1-t^2}{1+t^2},\quad dx=\frac{2}{1+t^2}dt}$$

使用范围：　　由三角函数和常数经过有限次四则运算构成的函数．一般记为

$$R(\sin x,\cos x)$$

如，$\displaystyle\int\frac{\sin x}{1+\sin x}dx$　　$\displaystyle\int\frac{1}{(2+\cos x)\sin x}dx$

例9 求积分 $\displaystyle\int \frac{\sin x}{1+\sin x+\cos x}dx.$

解 由万能代换公式 $\sin x = \dfrac{2t}{1+t^2}$,

$$\cos x = \frac{1-t^2}{1+t^2} \qquad dx = \frac{2}{1+t^2}dt,$$

$$\int \frac{\sin x}{1+\sin x+\cos x}dx = \int \frac{2t}{(1+t)(1+t^2)}dt$$

$$= \int \frac{2t+1+t^2-1-t^2}{(1+t)(1+t^2)}du$$

$$=\int\frac{(1+t)^2-(1+t^2)}{(1+t)(1+t^2)}dt \quad =\int\frac{1+t}{1+t^2}du \quad -\int\frac{1}{1+t}dt$$

$$=\arctan t \quad +\frac{1}{2}\ln(1+t^2) \quad -\ln|1+t|+C$$

$$\because\ t=\tan\frac{x}{2}$$

$$=\frac{x}{2} \ +\ln|\sec\frac{x}{2}|-\ln|1+\tan\frac{x}{2}|+C.$$

基本积分公式（续）

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(15) $\int \tan x dx = -\ln|\cos x| + C;$

(16) $\int \cot x dx = \ln|\sin x| + C;$

(17) $\int \sec x dx = \ln|\sec x + \tan x| + C;$

(18) $\int \csc x dx = \ln|\csc x - \cot x| + C;$

(19) $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C;$

$$(20)\quad \int \frac{1}{x^2-a^2}dx = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C;$$

$$(21)\quad \int \frac{1}{a^2-x^2}dx = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C;$$

$$(22)\quad \int \frac{1}{\sqrt{a^2-x^2}}dx = \arcsin\frac{x}{a} + C;$$

$$(23)\quad \int \frac{1}{\sqrt{x^2\pm a^2}}dx = \ln\left|x+\sqrt{x^2\pm a^2}\right| + C.$$

练习

1 $\int \frac{\mathrm{d}x}{1+\sqrt[3]{x+2}}$； 2. $\int \frac{1+\sin x}{3+\cos x}\mathrm{d}x$.

$$\int \frac{\mathrm{d}x}{1+\sqrt[3]{x+2}}.$$

解：令 $u=\sqrt[3]{x+2}$，则 $x=u^3-2,\ \mathrm{d}x=3u^2\,\mathrm{d}u$

$$\text{原式}=\int\frac{3u^2}{1+u}\,\mathrm{d}u \quad =3\int\frac{(u^2-1)+1}{1+u}\mathrm{d}u$$

$$=3\int(u-1+\frac{1}{1+u})\,\mathrm{d}u$$

$$=3[\ \tfrac{1}{2}u^2-u+\ln\left|1+u\right|\ ]+C$$

$$=\tfrac{3}{2}\sqrt[3]{(x+2)^2}-3\sqrt[3]{x+2}$$

$$+3\ln\left|1+\sqrt[3]{x+2}\right|+C$$

2. $\int \frac{1+\sin x}{3+\cos x}\mathrm{d}x.$

解 原式 $= \int \frac{1}{3+\cos x}\mathrm{d}x + \int \frac{\sin x}{3+\cos x}\mathrm{d}x$

前式令 $u=\tan\frac{x}{2}$；后式配元

$$= \int \frac{1}{3+\frac{1-u^2}{1+u^2}}\cdot\frac{2}{1+u^2}\mathrm{d}u - \int \frac{1}{3+\cos x}\mathrm{d}(3+\cos x)$$

$$= \int \frac{1}{u^2+2}\mathrm{d}u - \ln|3+\cos x|$$

$$= \frac{1}{\sqrt{2}}\arctan\frac{u}{\sqrt{2}} - \ln|3+\cos x| + C$$

$$= \frac{1}{\sqrt{2}}\arctan\left(\frac{1}{\sqrt{2}}\tan\frac{x}{2}\right) - \ln|3+\cos x| + C$$

小结

两类积分换元法：

$\begin{cases} \text{（一）凑微分} \\ \text{（二）三角代换、倒代换、根式代换} \end{cases}$

基本积分公式(续)

思考题

求积分 $\int (x \ln x)^p (\ln x + 1) dx.$

思考题解答

$$\because d(x\ln x)=(1+\ln x)dx$$

$$\therefore \int (x\ln x)^p(\ln x+1)dx=\int (x\ln x)^p d(x\ln x)$$

$$=\begin{cases}\dfrac{(x\ln x)^{p+1}}{p+1}+C, & p\neq -1\\ \ln\left|x\ln x\right|+C, & p=-1\end{cases}$$